EquipFlow
Phase 1 — Project Definition

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Purpose

This is the completed Phase 1 baseline for EquipFlow. It defines the product context, business problem, target users, equipment domain, knowledge base, core workflows, AI boundaries, tools, conceptual domain model, preliminary API surface, MVP scope, assumptions, constraints, risks, and success criteria.

It is a project-definition baseline, not a final architecture or implementation specification. Detailed requirements and technical choices will be produced in later phases.

Project	EquipFlow
Company	EquipTech Manufacturing
Domain	Industrial Manufacturing / Equipment Maintenance
Product	AI-powered Equipment Maintenance Assistant
Phase	1 — Project Definition
Status	COMPLETE — Baseline v1.0

1. Executive Summary

EquipFlow is an internal AI-powered equipment maintenance assistant for EquipTech Manufacturing. It helps maintenance technicians, engineers, and managers access trusted technical knowledge and operational maintenance information through a single interface.

The system combines retrieval-augmented generation (RAG) for documentation questions, controlled backend tools for operational data and actions, and a selective agentic workflow for genuinely multi-step maintenance analysis.

The MVP demonstrates a complete AI system: ingestion → retrieval → grounding → tools → selective agent reasoning → validation → citations/refusal → evaluation.

2. Project Identity

Item	Definition
Project Name	EquipFlow
Fictional Company	EquipTech Manufacturing
Company Type	Industrial manufacturing company
Product Type	AI-powered Equipment Maintenance Assistant
Primary Users	Maintenance Technician, Maintenance Engineer, Maintenance Manager
Primary Value	Fast, evidence-based access to maintenance knowledge and operational data

3. Business Problem

Maintenance information is distributed across equipment manuals, maintenance guides, troubleshooting guides, safety procedures, SOPs, and historical maintenance reports. Searching these sources manually is slow, fragmented, and makes it difficult to connect technical guidance with operational history.

Problem statement: EquipTech needs an internal assistant that can connect technical documentation with authorized maintenance data and provide evidence-backed answers without giving the AI uncontrolled access to business systems.

4. Proposed Solution

EquipFlow provides a controlled AI interface that selects the smallest appropriate execution path for each request.

```
Documentation question → RAG
Operational question → Tool
Combined question → RAG + Tool
Multi-step analysis → Maintenance Analysis Agent + Tools
Insufficient evidence → Refusal / clarification
```

The backend remains authoritative for authentication, authorization, validation, business rules, database access, and write operations.

5. Project Goals

- Provide fast access to equipment maintenance knowledge through natural-language interaction.
- Ground AI answers in the company knowledge base and provide source citations.
- Retrieve current operational maintenance data through controlled backend tools.
- Support combined documentation + operational-data questions.
- Demonstrate a meaningful multi-step agentic maintenance-analysis workflow.
- Prevent unsupported answers through evidence checks and refusal behavior.
- Create an evaluable AI pipeline rather than relying on subjective demos alone.
- Keep the MVP small enough to implement and test as a coherent system.

6. Target Users & Roles

Role	Responsibilities	Key capabilities
Maintenance Technician	Daily maintenance execution	Ask AI, search docs, view equipment/history, create reports
Maintenance Engineer	Diagnosis and analysis	Technician capabilities + deeper analysis and report review
Maintenance Manager	Oversight and administration	View fleet/history, review reports, upload/manage documents

Backend authorization enforces these permissions; the AI model is not an authorization authority.

7. Equipment Domain

Equipment Type	Example	Typical concerns
Industrial Pumps	P-101	Overheating, low pressure, leakage, cavitation
Electric Motors	M-101	Overheating, vibration, failure to start
Air Compressors	C-101	High temperature, low pressure, abnormal operation
Conveyor Systems	CV-101	Belt slippage, misalignment, motor failure

MVP scale: 3 production lines × 4 equipment types ≈ 12 equipment instances.

8. Knowledge Base / Documents

Document Type	Count	Purpose
Equipment Manuals	8	Specifications, operation, components, limits
Maintenance Guides	6	Maintenance and inspection procedures
Troubleshooting Guides	6	Symptoms, causes, diagnostics
Safety Procedures	4	Safety constraints and precautions
SOPs	4	Standard operating procedures
Maintenance Reports	4	Historical observations and outcomes
Total	32	Meets 30+ target

Formats: PDF and DOCX. Corpus: approximately 170–190 pages. Metadata includes ID, title, type, equipment, line, version, dates, and format.

9. Core User Workflows

9.1 Documentation / RAG

Example: “What are the common causes of P-101 overheating?”

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User → API → validation → AI orchestration → retrieval → relevant chunks → grounded generation → answer + citations
```

9.2 Operational / Tool

Example: “When was P-101 last serviced?”

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User → API → AI orchestration → maintenance-history tool → database → result → answer
```

9.3 Combined RAG + Tool

Example: “P-101 is overheating. What should I check, and when was its last maintenance?”

The system retrieves troubleshooting/safety evidence and separately queries maintenance history, then combines the evidence.

9.4 Agentic Workflow

Example: “Analyze why P-101 keeps overheating based on its maintenance history and the troubleshooting manuals.”

The Maintenance Analysis Agent may retrieve history, search documentation, compare findings, and produce a reasoned analysis through controlled tools.

9.5 Refusal

If sufficient evidence is unavailable, EquipFlow should refuse or ask for clarification instead of fabricating a plausible answer.

10. AI / RAG Concept

The MVP requires: document ingestion → parsing → extraction → chunking → metadata → embeddings → vector indexing → retrieval → context construction → grounded generation → citations → evaluation.

Hybrid retrieval and metadata filtering are required. Reranking is optional and will be justified by evaluation.

Evidence principle: the LLM is not treated as the authoritative source of company maintenance facts; it reasons over supplied evidence.

11. Agents & Tools Concept

Agent: Maintenance Analysis Agent — one MVP agent focused on multi-step maintenance analysis.

Agent boundaries: no direct SQL, unrestricted database access, authorization bypass, physical machine control, or invented evidence.

Tool	Purpose	Access
GetEquipment	Retrieve equipment details	Read
GetMaintenanceHistory	Retrieve maintenance events	Read
SearchMaintenanceReports	Find historical reports	Read
SearchKnowledgeBase	Retrieve relevant knowledge	Retrieval
CreateMaintenanceReport	Create validated record	Write + authorization

The model may request a tool call; the application decides whether it is permitted, validates it, executes it, and returns a controlled result.

12. Initial Conceptual Domain Model

Entities: User, ProductionLine, Equipment, MaintenanceEvent, MaintenanceReport, Document, DocumentChunk, AIConversation, AIMessage, AIExecution.

Relationships: ProductionLine → Equipment; Equipment → MaintenanceEvent; Equipment → MaintenanceReport; Document → DocumentChunk; AIConversation → AIMessage → AIExecution.

This is not the final database schema. Persistence, keys, indexes, cardinalities, and vector storage will be defined later.

13. Preliminary API Surface

Area	Endpoint	Purpose
Auth	POST /api/auth/login	Authenticate
Equipment	GET /api/equipment	List equipment
Equipment	GET /api/equipment/{id}	Details
Maintenance	GET /api/equipment/{id}/maintenance	History
Maintenance	POST /api/maintenance-reports	Create report
Documents	GET /api/documents	List/search
Documents	POST /api/documents	Upload
AI	POST /api/ai/chat	AI request
AI	GET /api/ai/conversations/{id}	Conversation

These are preliminary candidates, not final API contracts.

14. MVP Scope

In Scope

- Authentication and authorization with three roles.
- Equipment catalog for 3 production lines and approximately 12 equipment instances.
- PDF and DOCX document ingestion.
- 30+ documents and 150+ pages.
- Parsing, extraction, chunking, metadata, embeddings, and indexing.
- Hybrid retrieval and RAG-based question answering.
- Source citations and refusal behavior.
- Controlled maintenance-history and related backend tools.
- Combined RAG + tool workflows.
- One meaningful Maintenance Analysis Agent workflow.
- Maintenance report creation with validation and authorization.
- Basic AI execution tracing and evaluation.

Out of Scope

- Full CMMS or ERP functionality.
- IoT telemetry and real-time sensor ingestion.
- PLC integration or machine control.
- Automatic physical machine actions.
- Predictive-maintenance ML based on live sensor data.
- Native mobile applications.
- Complex scheduling/dispatch optimization.
- Enterprise-wide document lifecycle management.
- Autonomous safety-critical physical actions.
- Fine-tuning a foundation model as an MVP prerequisite.

15. Assumptions

- EquipTech owns or is authorized to use the documents.
- Maintenance records can be represented in structured application data.
- Users authenticate before protected operations.
- The system is internal.
- Documentation and authorized operational data are primary evidence sources.
- EquipFlow provides decision support and does not control physical equipment.

16. Constraints

- Strict MVP scope.
- Evidence availability limits answer reliability.
- AI output requires grounding and evaluation.
- Safety-critical physical actions remain outside the system.
- Architecture should remain practical for a small team.
- Technology choices must be justified by requirements.

17. Risks & Mitigations

Risk	Impact	Mitigation
Hallucination	High	Grounding, citations, evidence checks, refusal
Poor retrieval	High	Hybrid retrieval, metadata filtering, evaluation
Parsing errors	Medium/High	Ingestion validation and test corpus
Incorrect tool execution	High	Contracts, validation, authorization
Unauthorized access	High	Backend RBAC/authorization
Agent complexity	Medium	Agent only for genuine multi-step tasks
Scope creep	High	Strict MVP boundaries
Weak evaluation	High	Dedicated evaluation dataset

18. Success Criteria

- Documentation questions return grounded answers with citations.
- Authorized operational data is retrieved through controlled tools.
- Documentation and database evidence can be combined in one response.
- At least one realistic multi-step agent workflow works.
- The system refuses when evidence is insufficient.
- AI behavior is measurable with an evaluation set and traces.
- The solution stays within MVP scope.

19. High-Level System Concept

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User → Authentication/Authorization → ASP.NET API → AI Orchestrator → { RAG / Tools / Maintenance Analysis Agent }
→ Evidence/Data → LLM → Validation → Answer + Citations or Refusal
```

This is a conceptual boundary only. Final architecture, deployment, data stores, and technology choices are deferred.

20. Decisions Deferred

- Final functional and non-functional requirements.
- Detailed use cases and acceptance criteria.
- Final database schema and persistence strategy.
- Exact chunking and retrieval strategy.
- Vector database/search technology.
- LLM provider/model and embedding model.
- Agent framework/orchestration implementation.
- Clean Architecture/CQRS/MediatR adoption, if justified.
- Redis, background processing, observability, deployment, and cloud infrastructure.

21. Phase 1 Exit Criteria

Phase 1 is complete when the baseline is stable enough to drive Requirements Engineering:

- Product identity and business context defined.
- Problem and solution clearly stated.
- Users and responsibilities identified.
- Equipment and knowledge corpus bounded.
- Core AI workflows defined.
- RAG, tool, agent, and refusal boundaries explicit.
- Initial domain model and API surface identified without treating them as final.
- MVP and out-of-scope boundaries documented.
- Assumptions, constraints, risks, and success criteria recorded.
- Later-phase technical decisions explicitly deferred.

22. Phase 1 Status

Area	Status
Project Identity	COMPLETE
Business Context & Problem	COMPLETE
Solution & Goals	COMPLETE
Equipment & Knowledge Base	COMPLETE
Users & Roles	COMPLETE
Core Workflows	COMPLETE
RAG / Agents / Tools Concept	COMPLETE
Domain Model Direction	COMPLETE
Preliminary API Surface	COMPLETE
MVP Scope & Boundaries	COMPLETE
Assumptions / Constraints / Risks	COMPLETE
Success / Exit Criteria	COMPLETE

PHASE 1 — PROJECT DEFINITION: COMPLETE

Next: Phase 2 — Requirements Engineering. The next phase will translate this baseline into functional requirements, non-functional requirements, AI/RAG requirements, security requirements, evaluation requirements, use cases, and acceptance criteria.

Document control: Baseline v1.0. Future changes should be recorded as explicit project decisions and reflected in the next revision.